

Supervised Fine-Tuning

DATA 443/643 · Week 8, note 8.1

Week 8

American University

Ahmad Mousavi

Ideas from Stanford CS224N 2025 L10 (instruction finetuning as InstructGPT stage 1). Original slides; that course is not copied.

Lecture plan

- 1 Pretrain next-token, then follow a request
- 2 Mask the prompt
- 3 Quality beats noisy volume
- 4 The template is part of the method

Key goal: SFT copies answers. Preferences are Week 9.

First time: what / why / formula

- What / why. Pretrain already speaks. SFT teaches format: follow the user, answer as an assistant. Loss on the answer tokens, not the prompt. A base LM completes text; it does not reliably follow “Write a bullet list.” Preferences (Week 9) come after this.
- Architecture. Same transformer. Dataset of (prompt, answer). Causal mask unchanged. Label mask: prompt positions are -100 / ignored.
- Formula. $\mathcal{L} = - \sum_{t \in \text{answer}} \log p_{\theta}(y_t \mid y_{< t}, \text{prompt})$.
- Tradeoffs. + Fast behavior change. – Copies answers (including bad ones); can forget; not a preference model; noisy instruction data is method.

How. Collect or write pairs. Train a few epochs. Too many epochs: forget the base (note 8.2).

Same transformer, different data

- The base model already scores every continuation.
- CS224N L10: instruction FT is the first InstructGPT stage.
- Alpaca and Vicuna are this stage with public pairs.

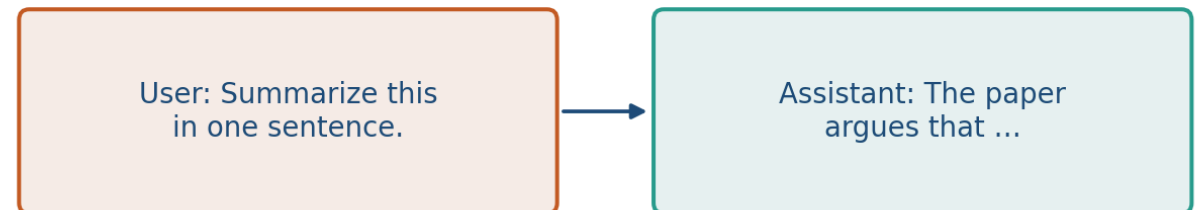
Instruction data is a distribution shift, not a new architecture



A row you would imitate

- Not a Wikipedia paragraph.
- A task plus an answer you would accept.
- System prompts and schemas live in the prompt, not a new net.

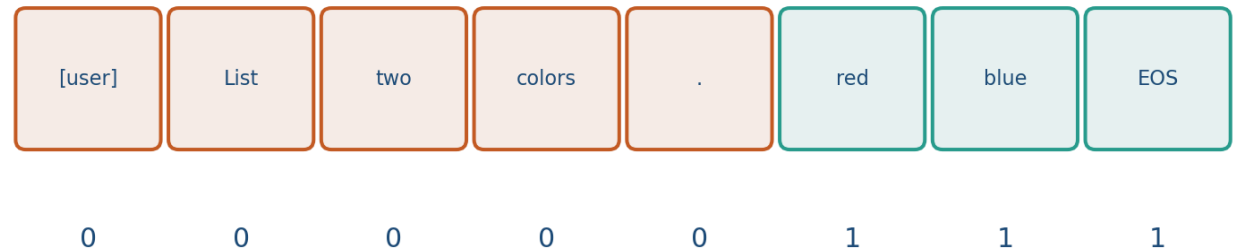
One SFT row: a request and a completion you would actually want



Zeros on the user

- Forward the prompt. Zero those positions in the loss.
- Otherwise the model spends gradient on imitating the user.
- Ten-token answers and 400-token answers are different datasets.

SFT mask: zeros on the prompt, ones on the response.



What should dominate the first run

80 clean medical pairs

Lead. Every response token is a label you want.

80,000 noisy threads

Token average lets the forum win 10:1. Filter or down-weight.

What to take from this lecture

Mask

Loss on Tr response tokens, not Tp + Tr.

Template

Train and eval with the same roles.

Not RLHF

SFT copies. Chosen-versus-rejected is Week 9.